

- Q1.** A doctor prescribes 2 pills per 10 pounds of body weight. If $f(x)$ represents the number of pills prescribed for a patient weighing x pounds, what is the value of $f(30)$?
 (A) 3
 (B) 4
 (C) 6
 (D) 8
 (E) 12
- Q2.** Which relation is a function?
 (A) A relation with two outputs for one input
 (B) A relation where each input has a unique output
 (C) A relation where each output has a unique input
 (D) A relation with no inputs
 (E) A relation with no outputs
- Q3.** What is the domain of the function $f(x) = \frac{1}{x}$?
 (A) All real numbers
 (B) All integers
 (C) All rational numbers except 0
 (D) All real numbers except 0
 (E) All complex numbers
- Q4.** A researcher studies the growth of bacteria in a controlled environment. The population P after t hours is modeled by $P(t) = 200 \cdot 2^{0.5t}$. What is $P(4)$?
 (A) 800
 (B) 1000
 (C) 1200
 (D) 1600
- (E) 2000
- Q5.** The vertical line test is used to determine if a relation is a
 (A) Function or not
 (B) Domain or range
 (C) Graph or table
 (D) Mapping or relation
 (E) Statistic or probability
- Q6.** If $g(x) = x^2 - 2x + 1$, what is $g(3)$?
 (A) 1
 (B) 2
 (C) 3
 (D) 4
 (E) 6
- Q7.** The range of the function $f(x) = |x|$ is
 (A) All real numbers
 (B) All non-negative real numbers
 (C) All integers
 (D) All rational numbers
 (E) All complex numbers
- Q8.** A patient's temperature T is modeled by $T(t) = 98.6 + 2\sin(\frac{\pi t}{12})$, where t is the time in hours. What is $T(6)$?
 (A) 98.6
 (B) 100.6
 (C) 100
 (D) 98
 (E) 96

Open Ended Questions (FRQ)

- Q9.** A hospital uses the function $C(x) = 2x + 5$ to calculate the cost of x doses of a medication. Find the domain and range of this function and explain the practical meaning of these values in the context of the hospital's medication costs.
- Q10.** A medical researcher collects data on the effect of a new medication on blood pressure. The data is modeled by the function $f(x) = 120 - 10 \cdot 2^{-x}$, where x is the number of hours after the medication is taken. Evaluate $f(2)$ and explain the practical meaning of this value in the context of the research.

Chef's Tips

- Tip 1: Ensure students can identify the domain and range from various representations
- Tip 2: Emphasize the importance of the vertical line test in determining functions
- Tip 3: Provide real-world applications to help students understand the practical significance of functions in healthcare and biology